

7. Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment. Make use of the `/proc//maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

```
GNU nano 2.7.1 practi
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <stdint.h>

int global_var = 100;
int uninitialized_global;
static int static_global = 200;

void display() {
    printf("Inside display() function\n");
}

int main() {
    int local_var = 10;
    static int static_local = 300;
    int *heap_var = malloc(sizeof(int));

    if (heap_var == NULL) {
        printf("Memory allocation failed\n");
        return 1;
    }

    *heap_var = 400;

    printf("Process ID: %d\n", getpid());
    printf("\n ----- Memory Addresses ----- \n");

    printf("Code address          : %p\n", (void *)(&display));
    printf("Initialized global         : %p\n", (void *)&global_var);
    printf("Uninitialized global       : %p\n", (void *)&uninitialized_global);
    printf("Static global              : %p\n", (void *)&static_global);
    printf("Static local               : %p\n", (void *)&static_local);
    printf("Heap variable              : %p\n", (void *)heap_var);
    printf("Stack variable             : %p\n", (void *)&local_var);

    printf("\nPress Enter to exit ... \n");
    getchar();

    free(heap_var);
    return 0;
}
```

```
thumu_hemanth_kumar@hemanths-TUF-Laptop:~$ cd OSPractical
thumu_hemanth_kumar@hemanths-TUF-Laptop:~/OSPractical$ nano practical7.c
thumu_hemanth_kumar@hemanths-TUF-Laptop:~/OSPractical$ gcc practical7.c -o practical7
thumu_hemanth_kumar@hemanths-TUF-Laptop:~/OSPractical$ ./practical7
Process ID: 797

----- Memory Addresses -----
Code address          : 0x5acf9ee50209
Initialized global     : 0x5acf9ee53010
Uninitialized global   : 0x5acf9ee53020
Static global          : 0x5acf9ee53014
Static local           : 0x5acf9ee53018
Heap variable          : 0x5acfaf3ec010
Stack variable         : 0x7fffd1b094bc

Press Enter to exit ...
^C
```